

# TINGFENG LAN

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## Research Interests

**Keywords:** LLM infra, storage systems for AI, serverless computing

- The overarching goal of my research is to enable fundamentally sustainable, scalable, and performant AI applications and platforms by exploring the intersection of AI and systems.
- Current research focuses: (1) rethinking the LLM storage backend through new system abstractions that better align with modern model and workload characteristics, and (2) cost-effective training and inference systems on modern and heterogeneous hardware platforms.

## Education

**University of Virginia**

Sep 2024 – Present

*Ph.D. in Computer Science, Advisor: Prof. Yue Cheng*

VA, USA

**Sichuan University**

Sep 2020 – Jun 2024

*B.Eng. in Computer Engineering, Advisor: Prof. Mingjie Tang*

Sichuan, China

## Industry Experience

**AntGroup AI Infra**

Sep 2023 – Jul 2024

Research Intern, Manager: Jian Sha

- Designed and implemented **DLRover-RM** (VLDB'24), a resource-aware optimization system for large-scale recommendation-model training that improves resource utilization and reduces training cost in cloud environments.
- Designed and implemented **m-LoRA** (VLDB'25), a multi-tenant LoRA training framework that enables parallel multi-adapter fine-tuning via pipeline parallelism, reducing memory redundancy and improving training throughput.

## Publications

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| <b>Preprint</b>    | <a href="#">Tingfeng Lan</a> , Zirui Wang, Zhaoyuan Su, Yunjia Zheng, Juncheng Yang, Yue Cheng. “ <b>TStore: Towards Practical Compression-Aware Model Storage with Tensor-Centric Fingerprinting and Clustering.</b> ”                                                                                                         |
| <b>Preprint</b>    | Yinghao Tang, <a href="#">Tingfeng Lan</a> , Xiuqi Huang, Hui Lu, Wei Chen. “ <b>SCORPIO: Serving the Right Requests at the Right Time for Heterogeneous SLOs in LLM Inference.</b> ”<br><i>In submission to IJCAI'26.</i>                                                                                                      |
| <b>Preprint</b>    | <a href="#">Tingfeng Lan</a> , Yusen Wu, Bin Ma, Zhaoyuan Su, Rui Yang, Tekin Bicer, Masahiro Tanaka, Olatunji Ruwase, Dong Li, Yue Cheng. “ <b>ZenFlow: Enabling Stall-Free Offloading Training via Asynchronous Updates.</b> ”<br><i>Adopted into <a href="#">DeepSpeed</a> and featured on <a href="#">PyTorch blog</a>.</i> |
| <b>ACL Demo'26</b> | Yinghao Tang, Yupeng Xie, Yingchaojie Feng, <a href="#">Tingfeng Lan</a> , Wei Chen. “ <b>IGenBench: Benchmarking the Reliability of Text-to-Infographic Generation.</b> ”                                                                                                                                                      |

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| ACL'26   | Yinghao Tang, Xueding Liu, Boyuan Zhang, <u>Tingfeng Lan</u> , Yupeng Xie, Jiale Lao, Yiyao Wang, Haoxuan Li, Tingting Gao, Bo Pan, Luoxuan Weng, Xiuqi Huang, Minfeng Zhu, Yingchao-jie Feng, Yuyu Luo, Wei Chen. <b>“IGenBench: Benchmarking the Reliability of Text-to-Infographic Generation.”</b>                                          |
| MLSys'26 | Zhaoyuan Su, <u>Tingfeng Lan</u> , Zirui Wang, Juncheng Yang, Yue Cheng. <b>“Efficient and Workload-Aware LLM Serving via Runtime Layer Swapping and KV Cache Resizing.”</b>                                                                                                                                                                    |
| MLSys'26 | Minchen Yu, Rui Yang, Chaobo Jia, Zhaoyuan Su, Sheng Yao, <u>Tingfeng Lan</u> , Yuchen Yang, Yue Cheng, Wei Wang, Ao Wang, Ruichuan Chen. <b>“λScale: Enabling Fast Scaling for Serverless Large Language Model Inference.”</b>                                                                                                                 |
| Preprint | Jiale Lao, Yinghao Tang, <u>Tingfeng Lan</u> , Mingjie Tang, Yuanchuan Zhou, Jianguo Wang. <b>“PathBee: Accelerating Shortest Path Querying via Graph Neural Networks.”</b><br><i>In submission to ICDE'26.</i>                                                                                                                                 |
| ICDE'26  | Ziling Huang, Zhengmao Ye, Qingsong Cai, Zelong Huang, Bo Sang, Haitao Zhang, Jian Sha, <u>Tingfeng Lan</u> , Hui Lu, Yuanchun Zhou, Mingjie Tang. <b>“DLRover-LM: LLM Pre-Training Framework with Thousands of Accelerators in AntGroup.”</b><br><i>In Proceedings of the 42nd IEEE International Conference on Data Engineering.</i>          |
| NSDI'26  | Zirui Wang, <u>Tingfeng Lan</u> , Zhaoyuan Su, Juncheng Yang, Yue Cheng. <b>“ZipLLM: Efficient LLM Storage via Model-Aware Synergistic Data Deduplication and Compression.”</b><br><i>In Proceedings of the 23rd USENIX Symposium on Networked Systems Design and Implementation.</i>                                                           |
| VLDB'25  | Zhengmao Ye*, Dengchun Li*, Zetao Hu, <u>Tingfeng Lan</u> , Jian Sha, Sicong Zhang, Lei Duan, Jie Zuo, Hui Lu, Yuanchun Zhou, Mingjie Tang. <b>“mLoRA: Fine-Tuning LoRA Adapters via Highly-Efficient Pipeline Parallelism in Multiple GPUs.”</b><br><i>In Proceedings of 51<sup>th</sup> International Conference on Very Large Data Bases</i> |
| VLDB'24  | Qinglong Wang*, <u>Tingfeng Lan</u> *, Yinghao Tang, Bo Sang, Haitao Zhang, Jian Sha, Hui Lu, Ke Zhang, Mingjie Tang. <b>“DLRover-RM: Resource Optimization for Deep Recommendation Models Training in the Cloud.”</b><br><i>In Proceedings of 50<sup>th</sup> International Conference on Very Large Data Bases</i>                            |

\* denotes equal contribution

## Open Source Projects

**DeepSpeed-ZenFlow: A stall-free offloading framework for LLM fine-tuning**

Oct 2024 - Present

Available on [DeepSpeed](#), Received 40k+ ★ on GitHub

- Designed and implemented **ZenFlow**, an importance-aware asynchronous offloading system that decouples GPU and CPU updates to eliminate GPU stalls. Achieved up to  $5\times$  end-to-end speedup,  $2\times$  reduction in PCIe traffic, and over 85% stall elimination while preserving accuracy.

## **mLoRA: A efficient multi-tenant LoRA training system**

Sep 2023 - May 2024

Received 300+ ★ on [GitHub](#)

- *Designed and implemented a training mechanism "BatchLoRA" which allows multiple LoRA adapters to share the pre-trained base model concurrently with reduced kernel launch overhead.*

## **DLRover: An efficient autotl system with fault-tolerance awareness**

Jun 2023 - March 2024

Received 1.6k+ ★ on [GitHub](#), Joined [LF AI & Data Foundation](#) ⚡

- *Designed and implemented a hyper-parameter autotuner to optimize performance-relevant configurations, like micro-batch size, for maximum hardware utilization. Achieved over 95% memory utilization within a 30s estimation and re-configuration time; An elastic trainer, allowing for real-time hyper-parameter configuration during training sessions, thereby eliminating the restart overheads typically necessary in conventional training frameworks.*

## **Funding and Grants**

2025 **Modal Research Grant**; Total: \$2,000 GPU credits; Led the proposal writing and submission;

## **Service & Activities**

2025-2026 **Shadow PC for EuroSys'26**

2025-2026 **Artifact Evaluation Committee for EuroSys'26, NSDI'26, OSDI'26**

2023–2024 **Journal Reviewer for IEEE TBD'24**

## **Talks**

2025 **ZenFlow: Enabling Stall-Free Offloading Training via Asynchronous Updates**;

Invited talk: DeepSpeed Team

2025 **ZenFlow: Enabling Stall-Free Offloading Training via Asynchronous Updates**;

Poster presentation: UVA 2025 AI/ML Resource Fair

## **Awards**

2026 **EuroSys Distinguished AEC**